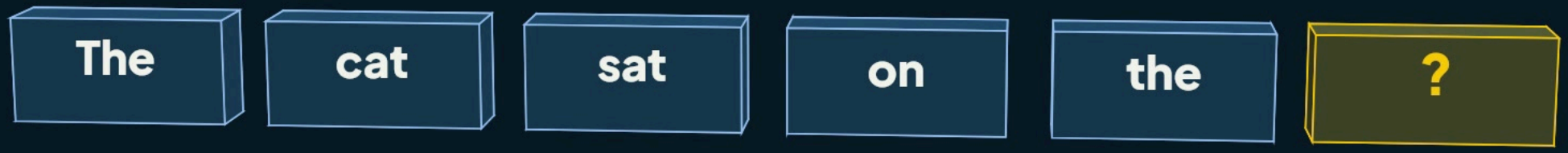
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A KINVECTA LEARNING SERIES

How Large Language Models Work



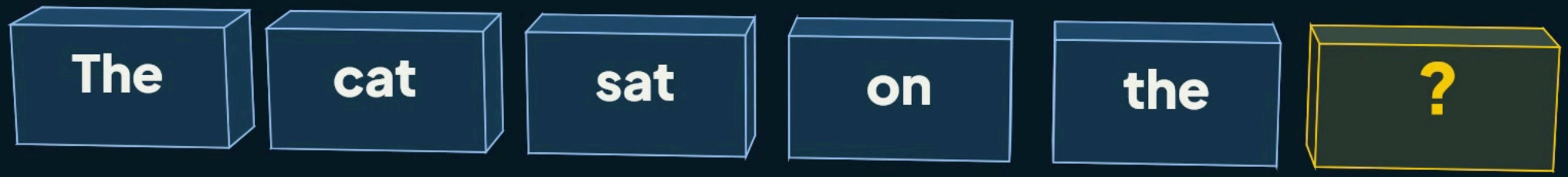
From math to transformers.

EPISODE 01 / WHAT AN LLM REALLY IS

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01 / A SENTENCE WITH A GAP

Five words. One gap.



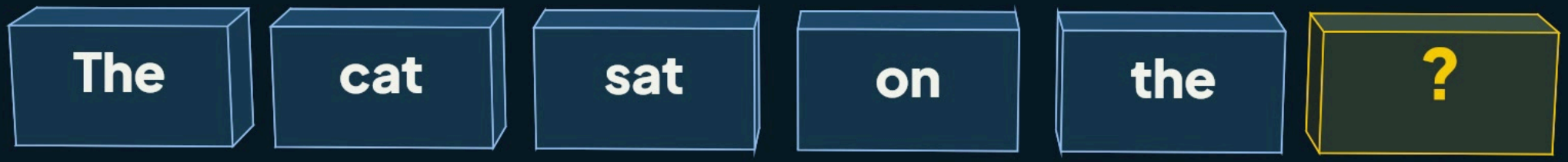
The text so far

SMALL LANGUAGE MODEL / RUNNING LOCALLY

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01 / A SENTENCE WITH A GAP

What comes next?



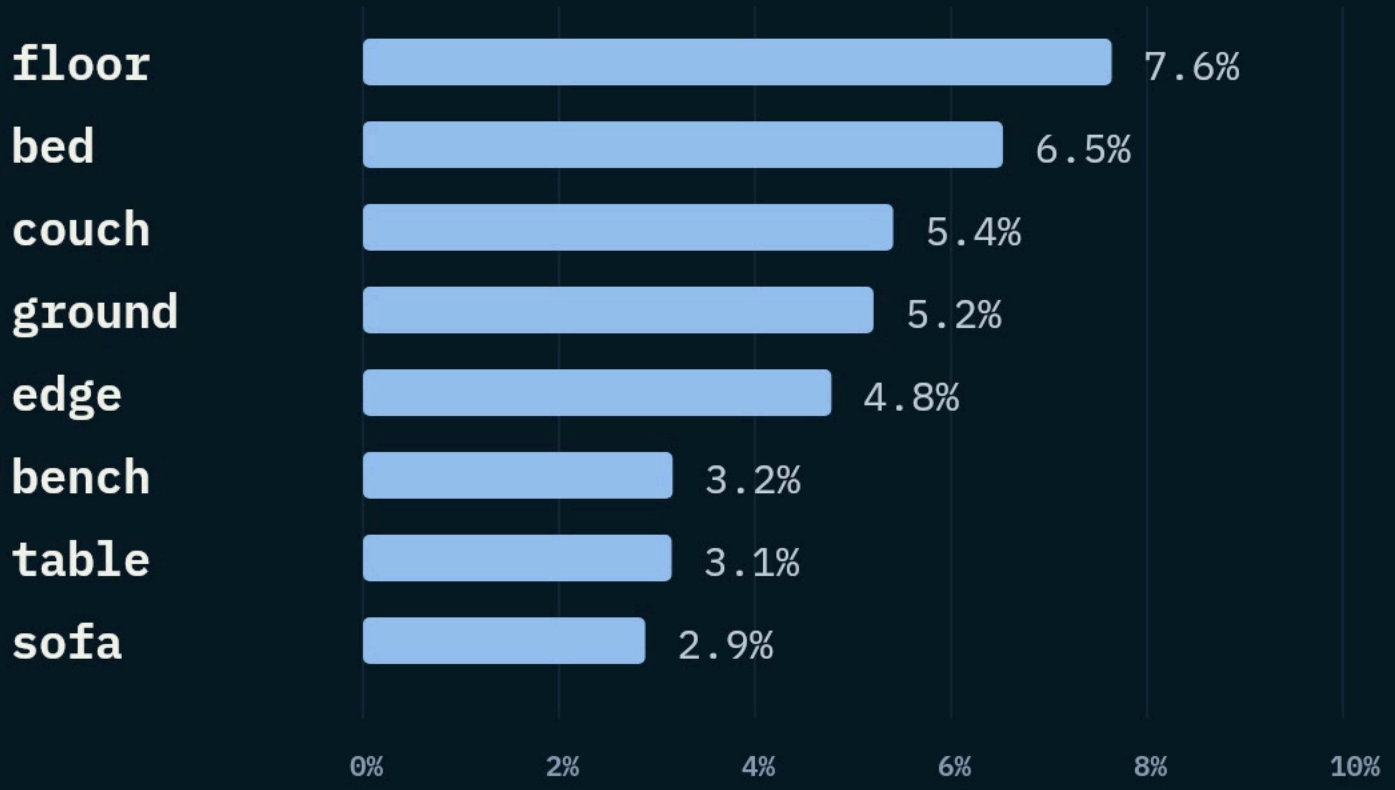
The text so far

SMALL LANGUAGE MODEL / RUNNING LOCALLY

02 / NEXT-TOKEN PREDICTION

A probability for every token.

GPT-2 SMALL / MEASURED OUTPUT



THE WORD YOU EXPECTED

mat

#31

in the model's ranking

0.40%

probability of "mat"

SAME PREFIX: The cat sat on the

03 / GENERATION

Pick. Append. Ask again.

ONE TOKEN JOINS THE PREFIX



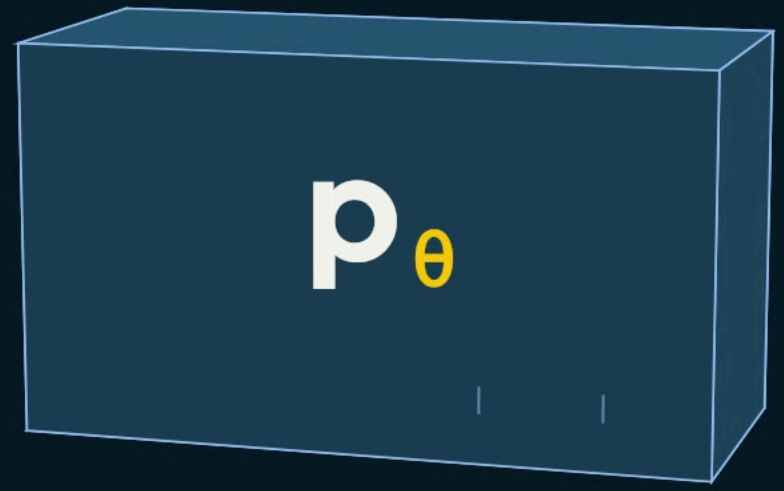
THE NEW PREFIX GOES BACK THROUGH THE MODEL

The cat sat on the floor,

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THE QUESTION THIS EPISODE ANSWERS

Where do those numbers come from?



Computation from learned parameters.

THE MODEL PRODUCES A DISTRIBUTION.

The cat sat on the…

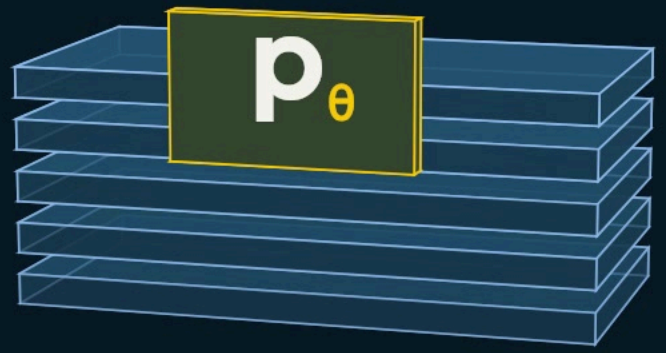
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HOW LARGE LANGUAGE MODELS WORK

From math to transformers.

EPISODE 01 / 30

What an LLM really is.



With Mark

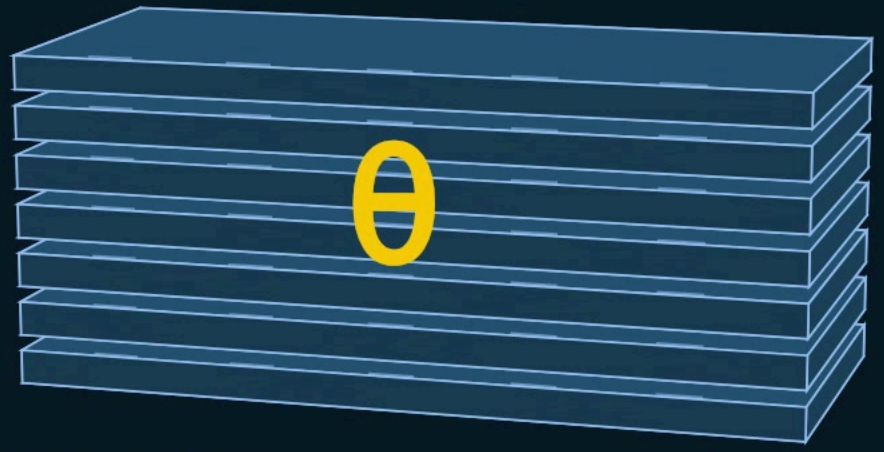
A FUNCTION. A DISTRIBUTION. A LOOP.

THE LITERAL MODEL

Learned weights + a small runtime.

LLAMA 2 / 70B / 16-BIT EXAMPLE

WEIGHTS



~140 GB

70 billion parameters

runtime

```
load weights θ

predict(prefix, θ)

sample(next_token)
```

~500 lines of C

Karpathy's compact example

SCHEMATIC: A MODEL PACKAGE MAY CONTAIN MORE THAN TWO PHYSICAL FILES.

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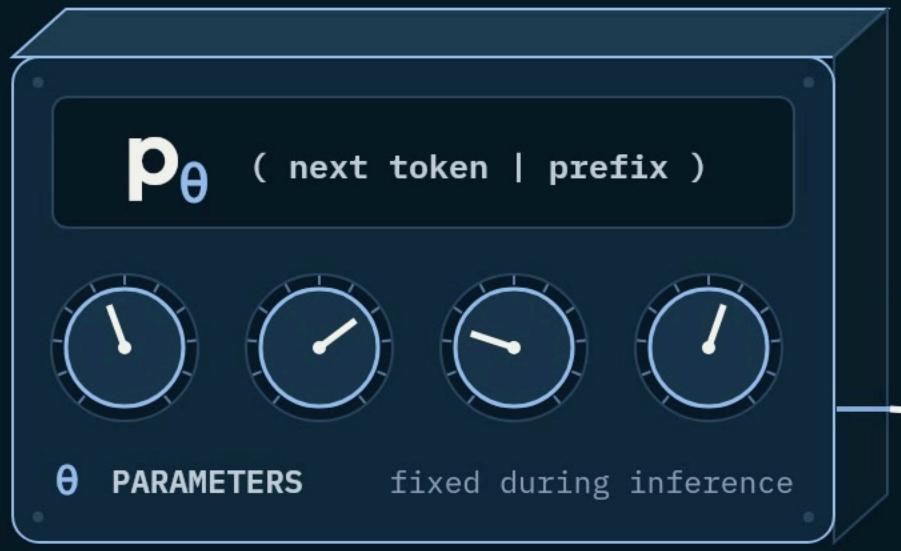
THE MODEL AS A FUNCTION

Inference uses the dials.

IN

The cat sat on the

text so far



OUT

floor	7.64%
bed	6.53%
couch	5.41%
ground	5.21%

+ 50,253 more probabilities

A function from the prefix to a distribution over tokens.

SCHEMATIC DIALS · GPT-2 SMALL OUTPUT SHOWN

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THE TWO WORDS WE NEED

Tokens and the vocabulary.

11 tokens

The	·cat	·sat	·on	·the	·mat	·because	·it	·was	·tired	.
464	3797	3332	319	262	2603	780	340	373	10032	13

TOKEN IDS SHOWN ABOVE
· MARKS THE ATTACHED SPACE

The period is a token, too.

Vocabulary

Every token the model can output.

50,257

tokens in V

GPT-2 SMALL TOKENIZER

One output probability for each token.

THE ONE EQUATION

The whole model, in one line.

$$p_\theta(x_t \mid x_{<t})$$

θ learned parameters

x_t next token

$x_{<t}$ the prefix



WATCH THE PREFIX GROW

New context. New probabilities.

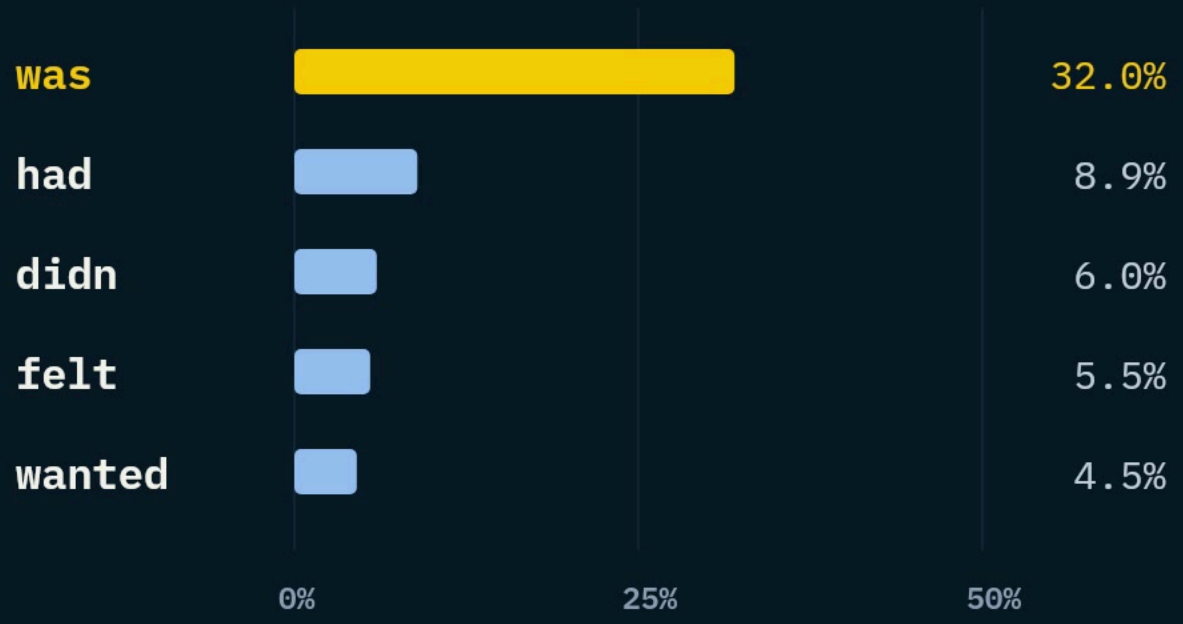
THE TEXT SO FAR

The cat sat on the mat because it

Next token

was

MEASURED GPT-2 SMALL OUTPUT

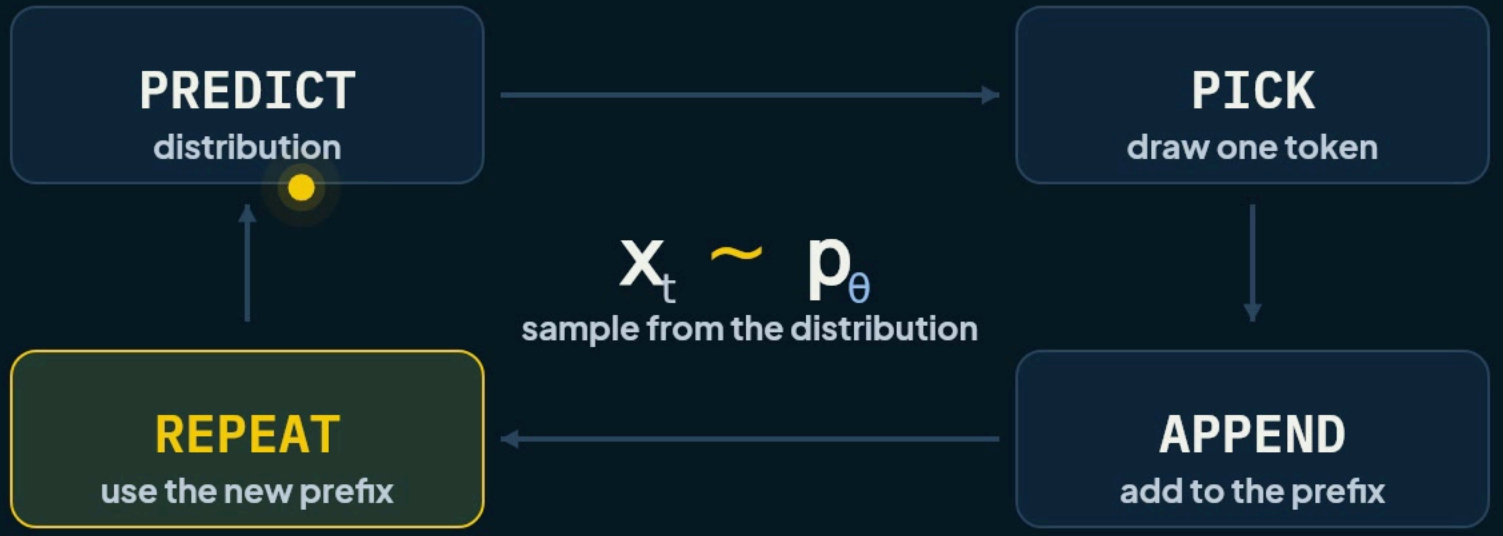


01 02 03 04

SELECTED PREFIXES FROM THE SAME SENTENCE · ALL 50,257 PROBABILITIES ARE RECOMPUTED.

AUTOREGRESSIVE GENERATION

One function. A growing sentence.



PREFIX

The cat sat on the floor ,

PREDICT → PICK → APPEND → REPEAT

SAME MODEL / SAMPLED EXCERPTS

Different draws. Different text.

PREFIX The cat sat on the

DRAW 01 … far side of the door with its head.

DRAW 02 … bench.

DRAW 03 … ceiling when it died, said Lew Gold.

SAME DIALS. SAME PREFIX. DIFFERENT RANDOM DRAWS.

MATH LAB / THE CHAIN RULE

Multiply the steps.

$$P(x_1, \ldots, x_t) = \prod_{i=1}^{t} p_\theta (x_i \mid x_{<i})$$

$$P(x_1) \times P(x_2 \mid x_1) \times P(x_3 \mid x_1, x_2) \times \ldots$$

OUR CAT EXAMPLE / GPT-2 SMALL

$$\approx 10^{-17}$$

One sentence among countless possibilities.

APPROXIMATE MEASURED EXAMPLE, CONDITIONED ON THE INITIAL TOKEN. THE FORMULA IS THE CHAIN RULE.

PREDICTION / CLAUDE SHANNON

Context makes the next letter easier.

HUMAN NEXT-LETTER GUESSING / ILLUSTRATION

T	H	E		C	A	T		S	A	T		O	N		T	H	E		?

1951

Claude Shannon

Guessing the next letter of English

0.6–1.3

bits per letter

Estimated with long context

A long prefix reduces uncertainty about what comes next.

WHAT BETTER PREDICTION REQUIRES

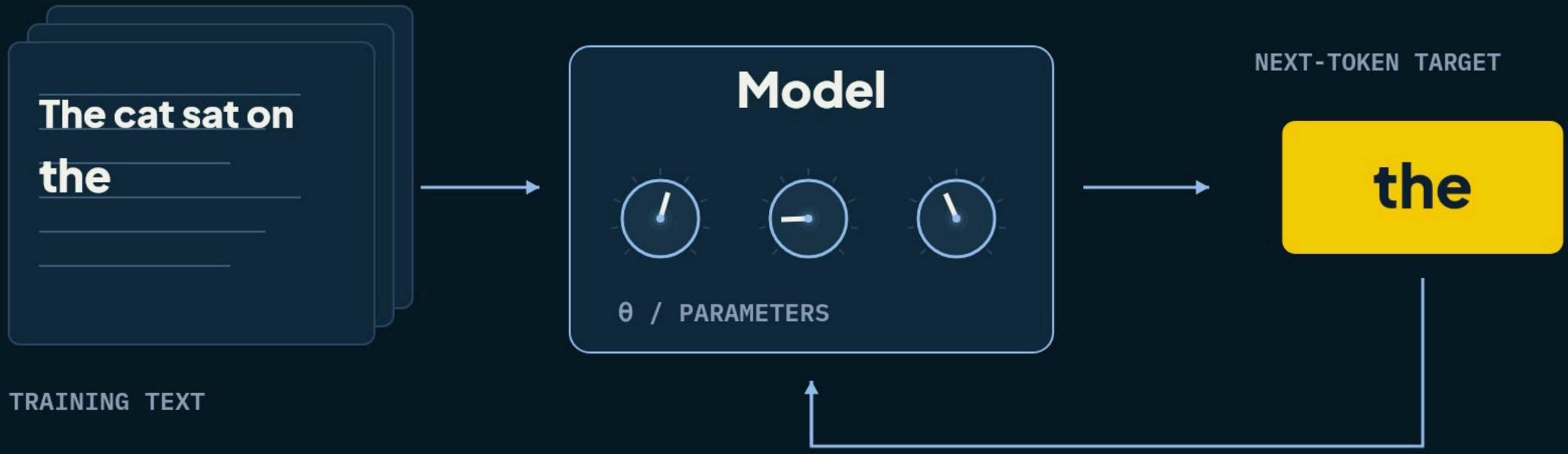
A small goal draws on many patterns.

GRAMMAR	The cat sat on	→	a noun phrase
FACTS	The capital of Texas is	→	Austin
STORY	The cat was tired.	→	It rested.
DIALOGUE	A question?	→	An answer.

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STAGE 01 / PRETRAINING

The text supplies its own targets.



Compare prediction with the next token in the text

ILLUSTRATION: TRAINING ADJUSTS θ; USING THE MODEL DOES NOT.

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STAGE 02 / POST-TRAINING

From continuing text to following requests.

INSTRUCTGPT / 2022 HISTORICAL EXAMPLE



People write good answers

~40 people / ~13k prompts

People rank the responses

4–9 candidates per prompt

Practice against the learned score

~31k more prompts

ONE INFLUENTIAL RECIPE / NOT A REQUIREMENT FOR EVERY MODEL

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NEWER STAGE / CHECKABLE PROBLEMS

Practice with checkable answers.

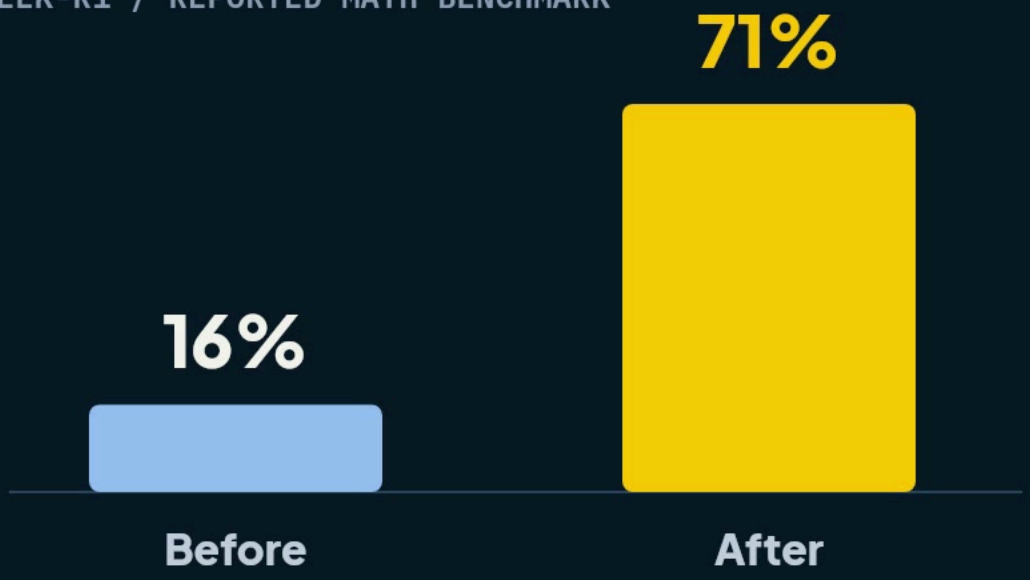
MATH AND CODE / SCHEMATIC TRAINING TASK

$$2 + 2 = 4$$



An answer can be checked.

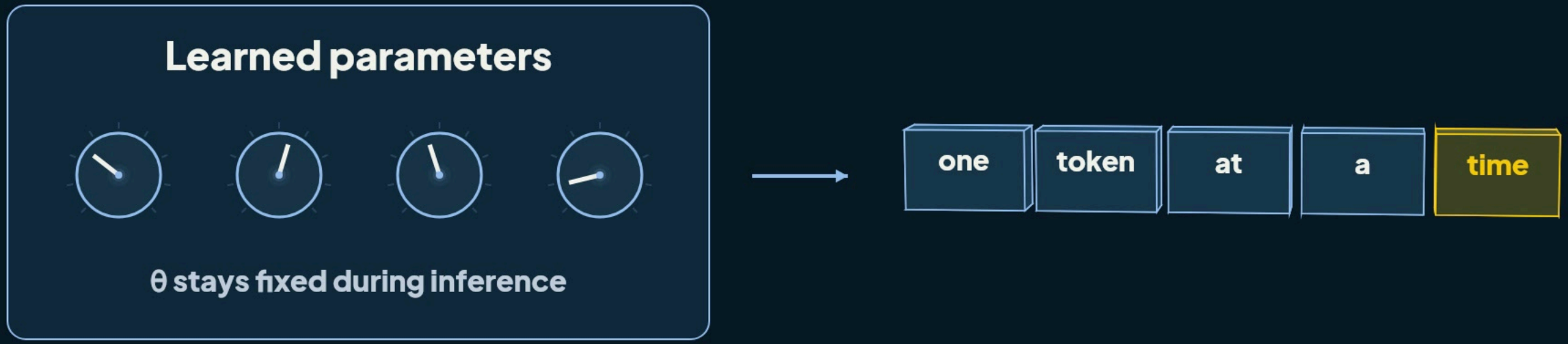
DEEPSEEK-R1 / REPORTED MATH BENCHMARK



2025 EXAMPLE / NOT A UNIVERSAL RECIPE OR A GENERAL ACCURACY SCORE

INFERENCE / USING THE FINISHED MODEL

Run the finished dials, one token at a time.



Prediction → pick → append → repeat

TRAINING CHANGES THE MODEL. INFERENCE USES IT.

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CLAIMS / CHECK BOTH SIDES

Three claims. Look at the evidence.

01 A database that copies text

02 It understands nothing

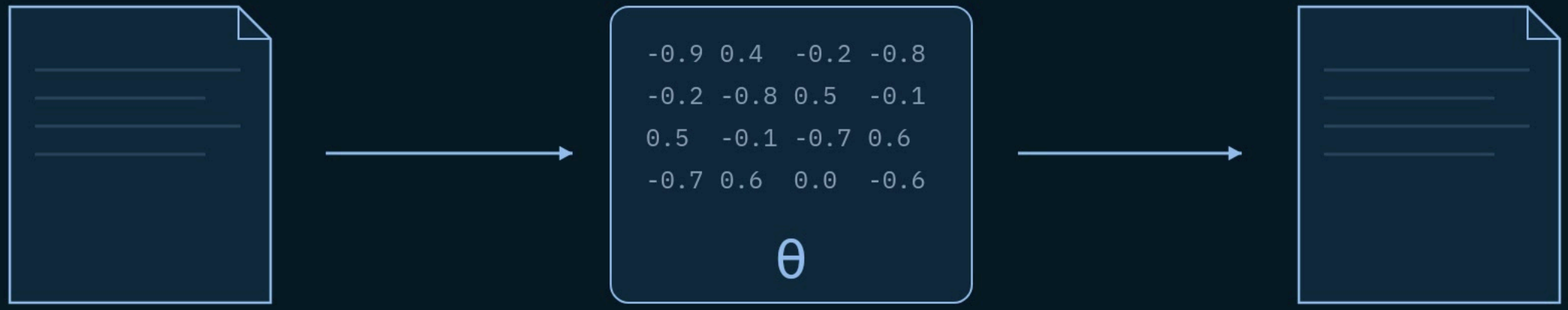
03 It thinks like a person

Evidence can support part of a claim and challenge another part.

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CLAIM 01 / COPYING

Memorization is real. Lookup is different.



MEMORIZATION Training can preserve exact sequences.

GENERATION A learned distribution, not a corpus search index.

EVIDENCE / OTHELLO-GPT

A board it was never shown.

TRAINING INPUT

Move sequences

f5 d6 c3 d3 c4

f4 f6 f3 e6 e7



Small GPT

ACTIVATIONS

FOLLOW-UP REPRESENTATION

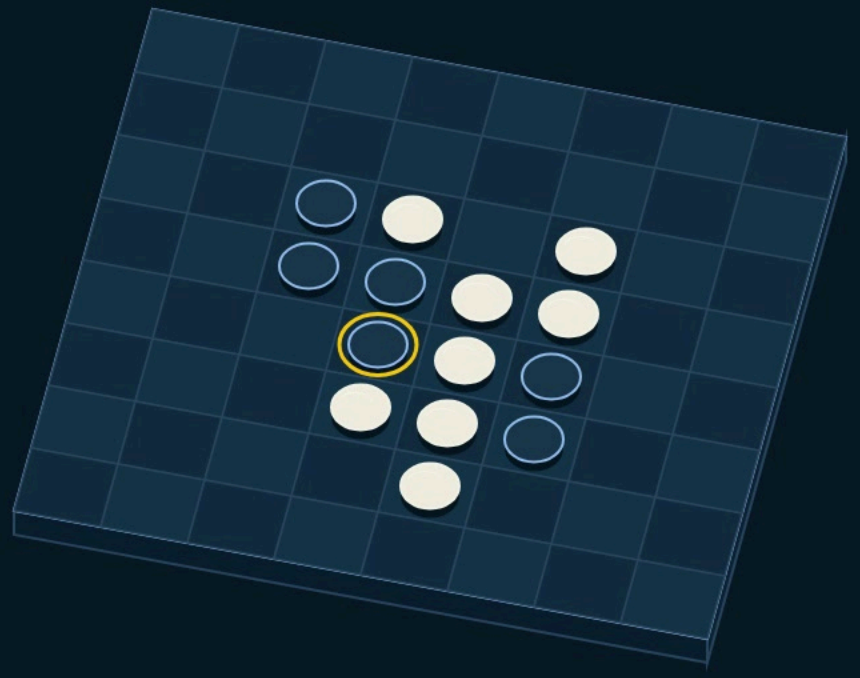
mine / yours

Edit the representation.
The next move changes.

Probe reconstruction error

Trained		1.7%
Untrained		~26%

PROBE READOUT



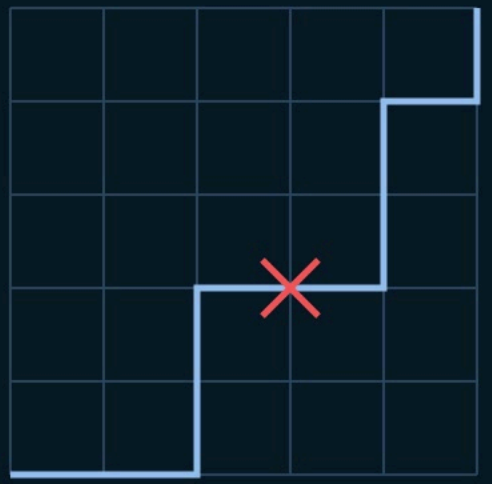
Illustrative representation edit

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COUNTER-EVIDENCE / THE TAXI MAP

Good predictions can hide a bad map.

Street grid



Detour

TURN SEQUENCES

S L R L R L

Implied map



Next-turn prediction

Brittle under detours

Schematic maps and route, not the study’s literal street reconstruction.

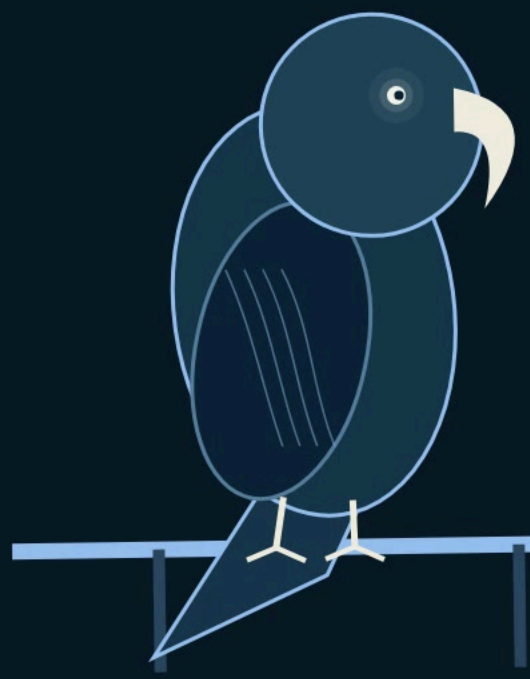
CLAIM 03 / THE STOCHASTIC PARROT

Fluent language. An open question.

hello

pretty bird

hello again



hello pretty bird

Patterns in language

The 2021 “stochastic parrot” argument

Internal structure

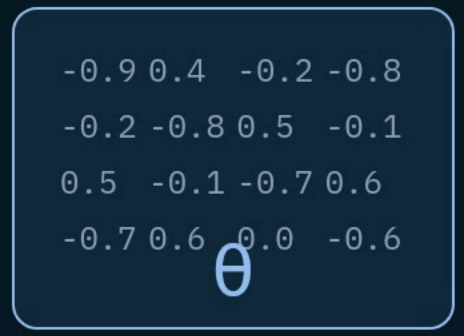
Othello is a reply.

Fluency alone does not settle understanding.

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THE ONE IDEA TO KEEP

A learned function, used in a loop.



Parameters + code



Prefix → probabilities



Predict · Pick
Append · Repeat

HOW THE DIALS ARE SET

Pretraining → Post-training → Reasoning RL*

* A newer route for some models

PAUSE & THINK

One question to take with you.

If a model is trained only to predict the next token, why might it still learn to answer questions?

Pause the video and work through your answer.

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ONE ANSWER

Questions often come with answers.

Question	→	Answer	A familiar continuation in text
Question	→	Answer	A familiar continuation in text
Question	→	Answer	A familiar continuation in text

Predicting the continuation rewards useful structure.

PREDICTION IS THE OBJECTIVE

NEXT / EPISODE 02

Vectors: words become numbers.

cat → [0.21, −0.47, 0.88, …]

ILLUSTRATIVE VECTOR



Companion notebook

Exact episode code



ILLUSTRATIVE PROJECTION / NOT MEASURED COORDINATES

Subscribe for Episode 2